

Project Design Phase

Problem – Solution Fit Template

Date	26 June 2026
Team ID	
Project Name	Metro Ticket Generating System
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ☐ Solve complex problems in a way that fits the state of your customers.
- ☐ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ☐ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ☐ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ☐ **Understand the existing situation in order to improve it for your target group.**

Metro Ticket Generating System – Design Thinking Analysis

Section	Content
1. Customer Segment(s)	• Metro Passengers • Metro Administrators • Ticket Inspectors • Customer Support Executives
2. Jobs-to-be-done / Problems	• Book metro tickets quickly • Calculate fare automatically • Generate digital QR tickets • Make secure online payments • View ticket history • Validate tickets efficiently • Reduce waiting time at ticket counters
3. Triggers	• Passenger plans to travel by metro • Passenger needs to book a ticket before boarding • Peak-hour travel requiring quick ticket booking • Administrator updates stations, routes, or fare rules • Ticket inspector verifies passenger tickets at entry/exit gates
4. Emotions (Before / After)	Before: Frustrated, stressed, worried about long queues, uncertain about fare and ticket availability. After: Confident, satisfied, informed, stress-free, and able to travel without delays.
5. Available Solutions	• Paper tickets at metro counters • Ticket vending machines • Smart cards • Mobile wallet payments • Manual fare enquiry at stations

Section	Content
6. Customer Constraints	• Long waiting queues • Manual ticket booking process • Limited payment options • Difficulty accessing real-time fare information • Lost or damaged paper tickets • Lack of centralized booking history
7. Behaviour	• Passenger visits the station or online portal • Selects source and destination stations • Checks fare • Makes payment • Downloads or displays QR ticket • Uses QR code at metro entry gate • Administrator monitors bookings and manages stations
8. Channels of Behaviour	Online: ServiceNow Service Portal, Mobile Browser, Email Notifications. Offline: Metro Ticket Counter, Ticket Vending Machine, Metro Entry/Exit QR Scanner.
9. Problem Root Cause	Existing ticket booking methods depend on manual counters, paper tickets, or isolated systems, resulting in long queues, delayed bookings, limited fare visibility, inefficient ticket validation, and lack of centralized ticket management.
10. Your Solution	Develop a Metro Ticket Generating System using ServiceNow , where passengers can log in, search metro routes, select source and destination stations, calculate fares automatically, make secure online payments, and receive QR-code-based digital tickets. Administrators can manage stations, routes, fare rules, and bookings through a centralized dashboard. Flow Designer automates booking confirmations, payment notifications, QR ticket generation, and email alerts, while role-based access control ensures secure and efficient system management.